

Early Blight of Tomato

NC STATE EXTENSION

Vegetable Pathology Factsheets

Pathogen

Early blight of tomato is caused primarily by the fungus *Alternaria linariae* (= *A. tomatophila*; formerly known as *A. solani*), although other species of *Alternaria* may be isolated from lesions.



Early blight symptoms on tomato.

Attribution: Inga Meadows



Early blight symptoms on tomato.

Attribution: Inga Meadows



Symptoms of early blight on tomato leaf

Attribution: Inga Meadows

Host Crops and Plants

Tomato is the most agriculturally important crop known to be affected by *Alternaria linariae*, although other members of Solanaceae and members of other plant families (Cucurbitaceae, Scrophulariaceae) can be affected.

Host Parts Affected

The leaves, fruit, and stem may be affected by *Alternaria linariae*.



Blighted tomato plants due to early blight.

Attribution: Inga Meadows

Identification

Symptoms of early blight can occur on the foliage, fruit, and stem at any stage of development. The disease is more commonly observed in the field, however, seedlings in the greenhouse can be affected by collar rot [<https://content.ces.ncsu.edu/collar-rot-and-alternaria-stem-canker-of-tomato/>] (also caused by species of *Alternaria*).

Lesions first develop on lower leaves as small, brownish-black spots which can expand to about $\frac{1}{4}$ - $\frac{1}{2}$ -inch in diameter with characteristic concentric rings in the darkened area. The area surrounding the lesions may become yellow and, as disease progresses, the entire leaf may turn yellow. In later stages, lesions may appear in the upper leaves and defoliation may occur in the lower part of the plant leaving the fruit susceptible to sunscald.

Fruit may become infected through the calyx around the stem attachment and are susceptible in the green or red stage. Lesions can expand to cover the entire fruit and are typically sunken, leathery, and dark brown to black with concentric rings.

Spores may be abundant on lesions on any part of the plant.

Look alike diseases: Several fungal diseases occur on the foliage of tomato and can make identification difficult. Septoria leaf spot [<https://content.ces.ncsu.edu/septoria-leaf-spot-of-tomato/>] produces smaller lesions with tan or light gray centers. Gray leaf spot also produces smaller and lighter brown lesions than what you would see with early blight, and the center of the gray leaf spot lesions tend to crack. Also, most hybrid tomatoes have resistance to gray leaf spot; however,

most, if not all, heirloom varieties lack this resistance. Late blight [<https://content.ces.ncsu.edu/tomato-late-blight>] produces lighter tan colored lesions that typically have a light green halo. Late blight will occur all over the plant including young leaves, whereas early blight would be found primarily on the lower leaves. Bacterial spot also may be confused with early blight when the early blight lesions are young and small or when both are present. In general, bacterial spot [<https://content.ces.ncsu.edu/bacterial-spot-of-pepper-and-tomato>] will be smaller (~1/16 in), may be water-soaked on the underside of the leaf, and the center of the lesion may fall out.



Early blight of tomato

Attribution: Inga Meadows



Septoria leaf spot on tomato leaf

Attribution: Inga Meadows



Gray leaf spot on tomato foliage

Attribution: Inga Meadows



Late blight on tomato foliage

Attribution: Inga Meadows



Water-soaked lesions of bacterial spot on tomato

Attribution: Erin Eure

Favorable Environmental Conditions for Disease

Disease is favored by warm temperatures (>80F) and moderate to heavy rainfall.

The fungus can survive between seasons on crop debris in the soil. Volunteer tomato plants also can be a primary source of inoculum.

Disease Transmission

Spores of *Alternaria linariae* are primarily spread by wind, air currents, and water splash. The fungus survives in plant debris in soil and, in milder climates, can survive on volunteer tomato plants.

General Disease Management

- Maintain plant vigor. Maintaining adequate fertilization and improving host vigor can reduce susceptibility to early blight.
- Remove volunteer weeds. Removing volunteer tomatoes can reduce disease as they can serve as an inoculum source.
- Rotate every 2-3 years. Rotating away from tomato can reduce inoculum density in a field.
- Apply fungicides preventatively.
- Plant tolerant varieties. There is no resistance to early blight, however, some varieties are more tolerant to the disease: 'Mountain Magic', 'Mountain Fresh Plus F1', 'Bush Celebrity Hybrid', 'Big Beef Hybrid', 'Celebrity Hybrid', 'Rutgers', 'Juliet F1', 'Tommy Toe', 'Old Brooks', or 'Cabernet F1'.

Disease Control for Conventional Growers

In addition to the cultural practices listed above, there are several products that are effective at controlling the disease. In general, apply on a 7-14 day interval except where the label specifies otherwise. For the latest fungicide recommendations for early blight, consult the [Southeastern US Vegetable Crop Handbook](https://content.ces.ncsu.edu/southeastern-us-vegetable-crop-handbook.pdf) [<https://content.ces.ncsu.edu/southeastern-us-vegetable-crop-handbook.pdf>].

Example products for control of early blight in tomato.

Active Ingredient	Example Product	PHI (days)	FRAC Group
azoxystrobin + chlorothalonil	Quadris Opti	0	11 + M
azoxystrobin + difenoconazole	Quadris Top	0	11 + 3
boscalid	Endura	0	7
chlorothalonil	Bravo	0	M
cyprodinil + fludioxinil	Switch	0	9 + 12
difenoconazole + benzovindiflupyr	Aprovia Top	0	7 + 3
difenoconazole + cyprodinil	Inspire Super	0	3 + 9
fenamidone	Reason	14	11
flutriafol	Rhyme	0	3
fluopyram + trifloxystrobin	Luna Sensation	3	7 + 11
fluopyram + pyrimethanil	Luna Tranquility	1	7 + 9
fluxapyroxad + pyraclostrobin	Priaxor	7	7 + 11
mandipropamid + difenoconazole	Revus Top	1	40 + 3
penthiopyrad	Fontelis	0	7
pydiflumetofen + fludioxonil	Miravis Prime	0	7 + 12
pyrimethanil	Scala	1	9
strobilurin ^R	Quadris, Cabrio, Gem, Evito	0	11
^R Resistance to this group of fungicides has been reported in North Carolina.			

Disease Control for Organic Growers

For organic control, the cultural control measures described above are the best means of control.

Some products are listed below, but consult the [Organic Materials Review Institute \(OMRI\)](http://www.omri.org/omri-lists) [<http://www.omri.org/omri-lists>] for a complete list of approved products.

Active Ingredient	Example Product
<i>Bacillus amyloliquefaciens</i> strain D747	Double Nickel 55
<i>Bacillus mycoides</i> isolate J	LifeGard WG
<i>Bacillus pumilus</i> strain QST 2808	Sonata
<i>Bacillus subtilis</i> strain IAB/BS03	AVIV
<i>Bacillus subtilis</i> strain QST 713	Cease, Serenad ASO, Serenade Opti
extract of <i>Reynoutria sachalinensis</i>	Regalia
fixed copper	various
polyoxin D zinc salt	OSO 5%SC
potassium bicarbonate	Carb-O-Nator
<i>Streptomyces lydicus</i> WYEC 108	Actinovate AG

Disease Control for Home Gardeners

In addition to the cultural practices mentioned above, home gardeners may reduce disease by the following.

- Trellis tomatoes to increase air flow and to prevent spores splashing up from the soil.
- Use a mulch to prevent spores from splashing up onto plants.
- Prune the bottom most leaves as the plant grows. These leaves are usually more infected than the upper parts of the plant.

Additional Resources

- The [NC State University Plant Disease and Insect Clinic](http://www.cals.ncsu.edu/plantpath/extension/clinic/) [<http://www.cals.ncsu.edu/plantpath/extension/clinic/>], provides diagnostics and control recommendations
- The [NC State Extension Plant Pathology Portal](http://plantpathology.ces.ncsu.edu/) [<http://plantpathology.ces.ncsu.edu/>] provides information on crop disease management
- The [Southeastern US Vegetable Crop Handbook](https://content.ces.ncsu.edu/southeastern-us-vegetable-crop-handbook.pdf) [<https://content.ces.ncsu.edu/southeastern-us-vegetable-crop-handbook.pdf>] provides information on vegetable disease management
- The [USDA Fungus-Host Distributions Database](https://nt.ars-grin.gov/fungaldatabases/fungushost/fungushost.cfm) [<https://nt.ars-grin.gov/fungaldatabases/fungushost/fungushost.cfm>] provides information about reported hosts for plant pathogenic fungi and oomycetes

Authors

Inga Meadows

Extension Associate, Vegetable and Herbaceous Ornamental Pathology Entomology & Plant Pathology

Ella Hinchliffe

Temporary Research Specialist Entomology & Plant Pathology

Sara Villani

Assistant Professor and Extension Specialist-Fruit and Ornamental Pathology Entomology & Plant Pathology

Publication date: Dec. 11, 2015

Reviewed/Revised: April 24, 2025

The use of brand names in this publication does not imply endorsement by NC State University or N.C. A&T State University of the products or services named nor discrimination against similar products or services not mentioned.

Recommendations for the use of agricultural chemicals are included in this publication as a convenience to the reader. The use of brand names and any mention or listing of commercial products or services in this publication does not imply endorsement by NC State University or N.C. A&T State University nor discrimination against similar products or services not mentioned. Individuals who use agricultural chemicals are responsible for ensuring that the intended use complies with current regulations and conforms to the product label. Be sure to obtain current information about usage regulations and examine a current product label before applying any chemical. For assistance, contact your local N.C. Cooperative Extension county center.

N.C. Cooperative Extension prohibits discrimination and harassment regardless of age, color, disability, family and marital status, gender identity, national origin, political beliefs, race, religion, sex (including pregnancy), sexual orientation and veteran status.

This publication printed on: Aug. 10, 2026

URL of this page [<http://content.ces.ncsu.edu/early-blight-of-tomato>]

